AMC 2026 Round 2 – Cheat sheet

Financial Modeling (USI) · TA Session 4 · goes with the notebook `02_round2_lecture.ipynb` (§*n* = notebook section)

The numbers of the TA example come from the CSV files on iCorsi (prices up to 26 Oct 2026, rates up to 23 Oct 2026). If you download newer data, your numbers will be slightly different.

What your Round 2 report must contain

1. **Stock analysis:** valuation, specific issues, catalysts. The catalysts are the central part.
2. **Round 1 performance:** return, volatility and Sharpe Ratio vs the 8% target, plus a performance attribution.
3. **Checkpoint 1** (alphas) and **Checkpoint 2** (valuation dossier), each graded complete/incomplete.
4. **Portfolio description:** positions, identifiers, number of shares, weights, price date, and the ex-ante TE vs the SAA.

§0 The data (notebook 01_download_data)

The notebook `01_download_data.ipynb` downloads the data needed for the project and saves them as CSV files.

File	What	Manipulation	Used for
`r2_adj_prices.csv`	Daily prices of the 8 SAA ETFs and the 3 stocks, adjusted for dividends and splits, from 2016	Month-end prices, then monthly returns minus the T-bill	Returns, beta, residual risk, TE, vol, VaR (§1–§2, §6–§8)
`r2_close_prices.csv`	Same, raw closing prices (not adjusted)	Last available close	P_0 for the valuation and the number of shares (§3–§4, §10)
`r2_dividends.csv`	Dividends per share of the 3 stocks, one row per payment	Sum of the last 12 months	D_0 for the DDM (§3–§4)
`r2_rates.csv`	1-month T-bill and 10-year Treasury yield, % per year	Divided by 100; T-bill divided by 12 for the monthly rate	T-bill: excess returns and Sharpe. 10-year: CAPM k (§3)
`r2_fundamentals.csv`	Annual income statement, balance sheet and cash flow of the 3 stocks and their competitors	Ratios: ROE, FCF margin, revenue growth, gross margin, debt/equity, payout/FCF	Value-trap check (Checkpoint 2)
`r2_multiples.csv`	Forward EPS, trailing EPS and forward P/E of the 3 stocks and of their competitors	Average forward P/E of the competitors; forward vs trailing EPS	Relative valuation (§4), value-trap check (Checkpoint 2)

§1 Monthly returns

$$R_t = \frac{P_t}{P_{t-1}} - 1, \qquad R_t^e = R_t - r_{f,t}, \qquad \text{last 60 months}$$

§2 The SAA is the benchmark, the passive portfolio and the funding

$$R_{S,t} = \sum_j w_j^S R_{j,t}, \qquad \sigma_S = \text{sd}(R_S)\sqrt{12}, \qquad \text{VaR}_{5\%,1m} = 1.645\,\frac{\sigma}{\sqrt{12}}, \qquad SR_S = \frac{E(R_S^e)}{\sigma_S}$$

§3 Inputs and cost of equity (CAPM)

$$k = r_f + \beta\,\text{ERP}, \qquad \beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

Example (Class 5): KO, $k = 4.4\% + 0.616 \times 6.114\% = 8.166\%$.

What k is. The cost of equity: the return shareholders require to hold the stock, given its risk. The CAPM builds it in two pieces: the risk-free rate (the compensation for time) plus a risk premium (the compensation for risk). The risk premium is the ERP (the extra return the stock market pays over the risk-free rate; 6.114% from Bloomberg CRP in Class 5, the same for all your stocks), scaled by the stock's beta: a stock with beta 1.2 moves 20% more than the market, so it must pay 20% more of the ERP.

What it is used for.

- In the DDM (§4) it is the **discount rate**: future dividends are worth less today, and more so the riskier the stock.
- In the alpha (§5) it is the **benchmark**: alpha is the expected return above k. A stock is attractive only if you expect it to earn more than its risk requires.

Which risk-free rate: 10-year Treasury or 1-month T-bill?

	1-month T-bill	10-year Treasury
Maturity	1 month	10 years
Used for	Monthly excess returns, the Sharpe Ratio of the challenge, the regressions in §6	r_f in the CAPM cost of equity k
Why	Matches the **monthly** holding period of the returns we measure	Matches the **long horizon** of the cash flows we discount: a DDM discounts dividends for decades, so the discount rate should be a long-term rate

The rule: match the maturity of the risk-free rate to the horizon of what you compare it with. In Class 5 the 10-year yield was 4.4%.

Inputs = information available today: the last dividend, analyst forecasts of g, beta from past returns. Nothing from the future.

The inputs, one by one

- P_0: the last closing price, not adjusted for dividends. In the challenge use the close of Fri 6 Nov.
- D_0: the dividends per share paid over the last 12 months. The first dividend you discount is $D_1 = D_0(1+g)$.
- β: the slope of the stock's monthly returns on the market (SPY), last 60 months, as in the formula above. It measures systematic risk only, the part you are paid for. It is **not** the beta on the SAA of §6.
- k: the CAPM above. Example, Nike with the data of the TA example:

$$k_{NKE} = 5.00\% + 1.031 \times 6.114\% = 11.30\%$$

- g_1, g_2, T: g_1 is the growth of the dividend in the first phase, g_2 the growth forever after, T the number of years of the first phase. g_1 can be **taken from Bloomberg**: analysts' consensus growth forecast (`EE`, or the dividend forecasts in `BDVD`). g_2 and T are **not in Bloomberg**: they are your assumptions. Keep g_2 below k and close to long-run economic growth, choose T as the years you expect the company to need to reach its normal state, and justify both in the report.

> **Careful:** if the company returns a large part of its cash through **buybacks** (e.g. NKE until FY2025, when buybacks were larger than dividends: see the line `Repurchase Of Capital Stock` in `r2_fundamentals.csv`), dividends understate the payout and the DDM undervalues the stock: use dividends + buybacks per share, or comparables.

Gordon or two-stage? Gordon assumes the dividend grows at the same rate forever, so use it for a mature company with stable growth (GIS in the notebook). Use two-stage when the company is not in its normal state today, for example a turnaround (NKE) or earnings at a cyclical low (HD): higher growth g_1 for T years while it recovers, then a normal g_2. One constant g would either ignore the recovery or extrapolate it forever.

§4 Intrinsic value

Gordon (constant growth, requires $k > g$):

$$V_0 = \frac{D_0(1+g)}{k-g}$$

Two-stage (g_1 for T years, then g_2 forever):

$$V_0 = \sum_{t=1}^{T} \frac{D_0(1+g_1)^t}{(1+k)^t} + \frac{1}{(1+k)^T} \cdot \frac{D_T(1+g_2)}{k-g_2}$$

Margin of safety at the current price:

$$\text{MoS} = \frac{V_0}{P_0} - 1$$

The value V_0 is an estimate built on k and g, so buy only if the stock stays cheap even when your inputs are a bit too optimistic. Small errors matter because $k - g$ is small: for KO, $g = 5.9\%$ gives USD 95.33 (+37% vs USD 69.52), while $g_2 = 5.0\%$ gives USD 72.93 (+5%). **Report a sensitivity table on g_2 and k.**

> **Red flag:** an upside above about +50% on a large, mature company is more likely a wrong input or a value trap than a mispricing. Check the inputs and the quality screen first.

Relative valuation (comparables). Price the stock at the average multiple of its competitors in the same industry (Class 5):

$$P_i = EPS_i \times \overline{P/E}_{\text{comparables}}$$

Use next year's consensus EPS (Bloomberg `EE`) and the forward P/E of each competitor (Bloomberg `RV`, which lists the peers with their multiples side by side). P/B or P/S work the same way. It is a second opinion next to the DDM: if both say the stock is cheap, the case is stronger. It is especially useful when the DDM fits badly, for example when the company pays out a large part of its cash through buybacks. Comparables used in the notebook: NKE vs LULU, DECK, Adidas; HD vs LOW, FND, TSCO; GIS vs KHC, CAG, CPB.

Example: Nike (NKE), data of the TA example (the NKE row of the notebook)

Data (USD): $P_0 = 37.91$, $D_0 = 1.64$, $\beta = 1.031$, $r_f = 5.00\%$ (10-year), ERP $= 6.114\%$, $g_1 = 10\%$ for $T = 5$ years, $g_2 = 4\%$.

Step 1: cost of equity

$$k = 5.00\% + 1.031 \times 6.114\% = 11.30\%$$

Step 2: Gordon (one growth rate, $g = 4\%$)

$$V_0 = \frac{D_0(1+g)}{k-g} = \frac{1.64 \times 1.04}{0.1130 - 0.04} = \frac{1.706}{0.0730} = 23.36$$

Step 3: two-stage

$$V_0 = \sum_{t=1}^{5} \frac{1.64 \times 1.10^t}{1.1130^t} + \frac{1}{1.1130^5} \cdot \frac{2.641 \times 1.04}{0.1130 - 0.04}$$

t	$D_t = 1.64 \times 1.10^t$	$(1.1130)^t$	$D_t/(1.1130)^t$
1	1.804	1.113	1.621
2	1.984	1.239	1.602
3	2.183	1.379	1.583
4	2.401	1.535	1.565
5	2.641	1.708	1.546
		Sum	**7.92**

Terminal value at year 5, then discounted to today:

$$TV_5 = \frac{2.641 \times 1.04}{0.1130 - 0.04} = 37.63, \qquad \frac{37.63}{1.708} = 22.03, \qquad V_0 = 7.92 + 22.03 = 29.95$$

(The notebook gives 29.94 because it does not round k.)

Step 4: compare with the price

$$\text{MoS}_{\text{Gordon}} = \frac{23.36}{37.91} - 1 = -38\%, \qquad \text{MoS}_{\text{2-stage}} = \frac{29.95}{37.91} - 1 = -21\%$$

Both values are below P_0: with these inputs NKE looks **overvalued**, so the signal is to sell or short, not to buy. Even the optimistic two-stage (10% growth for 5 years) does not justify the price. In the challenge portfolio this means $\alpha < 0$, so weight 0.

§5 Alpha: why catalysts matter

Class 5 gives two cases, both over a **1-year horizon**:

$$\alpha^{\text{closed}} = (1+k) \cdot \frac{V_0 - P_0}{P_0} \quad \text{(KO: 40.1\%)}, \qquad \alpha^{\text{never}} = \frac{D_1}{P_0} + g - k \quad \text{(KO: 0.84\%)}$$

Your judgment (justify it in the dossier), for each stock:

- **Catalyst** → the gap closes within 1 year → use α^{closed}.
- **No catalyst**, or the value-trap check fails → the gap never closes → use α^{never}, or drop the stock.

No catalyst means almost no alpha, however big the gap.

Horizon: the alpha is annual, like β_i and $\sigma^2(\varepsilon_i)$ in §6, so Treynor-Black compares quantities on the same scale. Over the 3 months of the challenge the expected alpha is roughly $\alpha/4$.

"A historical average is not an alpha." Your alpha comes from the valuation **and** the catalyst.

In the notebook: NKE and HD have a negative gap (weight 0 anyway). GIS has a +143% gap, but it fails the value-trap check, so the gap does not close: $\alpha_{GIS} = \alpha^{\text{never}} = 4.35\%$ (instead of 151%).

Where to find alphas and catalysts

Finding the mispricing (the source of α):

Idea	How
Value screen	P/E, P/B or EV/EBITDA below peers or below the stock's own history (Bloomberg `EQS`, Yahoo or Finviz screeners)
Value + quality	Keep only cheap stocks with stable margins, high ROE and strong FCF, to avoid the value trap
Reverse the DDM	Solve for the growth the price implies, and compare it with your own forecast
Overreaction	A big drop after news that doesn't change the long-run fundamentals
Analyst targets	Consensus target vs price (Bloomberg `ANR`), a starting point to investigate, not an alpha by itself

Finding the catalyst (between 6 Nov and the end of December):

Catalyst	Where to look
Earnings releases and guidance	Bloomberg `ERN` / `EVTS`, Yahoo earnings calendar, company IR page
Investor day, new strategic plan, new CEO	Company IR page, news
Buyback or dividend increase	Company press releases
Spin-off, M&A, asset sale	News, company filings
Activist investor stake	SEC EDGAR (13D filings); activist investing
Index inclusion	S&P index announcements (the quarterly rebalance takes effect in December)
Regulatory or legal decision	Agency calendars (e.g. FDA decision dates), court schedules

Checkpoint 2: the valuation dossier

For each stock: **1. method** (and which methods don't fit), **2. valuation** (V_0 with all inputs, margin of safety), **3. value-trap check**, **4. catalyst** before the end of the challenge.

The value-trap check. A cheap stock is either **mispriced** (the gap will close) or a **value trap**: it is cheap because its earnings are about to fall, so the low P/E corrects through E going down, not P going up (Class 5).

When: only for stocks that look cheap, i.e. **positive alpha**. A negative alpha means the stock is expensive: in theory you would short it, but Round 2 is long-only, so its weight is 0 and there is nothing to check.

1. Compute 6 metrics, one or two for each quality characteristic of Class 5:

$$\frac{V}{B} = \frac{\text{ROE} \times \text{payout ratio}}{k - g}$$

Characteristic	Metric	Good if
Profitability	ROE = net income / equity; FCF margin = FCF / revenue	high (the business generates cash)
Growth	revenue growth; gross margin	high (falling margin = growth bought with price cuts)
Safety	debt / equity	low
Payout	(dividends + buybacks) / FCF	low (> 1 = paid with debt)

FCF = operating cash flow − capital expenditure. With negative equity ROE and D/E are meaningless. FCF margin is defined even after a loss.

2. Check each metric on two dimensions (last fiscal year):

- **Level:** vs the median of the competitors (the same peers as the relative valuation). *Is the business weaker than its industry?*
- **Trend:** vs the stock's own average of the previous years. *Is it getting worse?*

Worse than peers **and** worsening = **red flag**.

3. Check expected earnings: forward EPS (analysts' forecast for the next 12 months) vs trailing EPS (last 12 months). If forward $<$ trailing, analysts expect earnings to fall and the low price may be right.

> **Careful:** with one-off items: a trailing loss caused by a write-down makes any forward EPS look like a rise.

Verdict: red flags + falling earnings → value trap: assume the gap does not close (§5) or drop the stock. Management quality (acquisitions, governance) is your judgment from the 10-K and the earnings calls.

In the notebook (GIS): red flags on revenue growth and debt/equity; the EPS "rise" ($-1.64 \to 3.15$) only reflects the one-time loss of the last 12 months (in FY2024–FY2025 EPS was 4.1–4.3). Cheap for a reason → value-trap risk → α^{never}.

Checkpoint 1: the alphas

One page:

1. **3–6 stocks with a positive alpha**, each with its justification (valuation + catalyst). If your valuation says a stock is overvalued, its weight is 0: replace it.
2. **Weights** proportional to $\alpha_i/\sigma^2(\varepsilon_i)$, with $\sigma^2(\varepsilon_i)$ estimated on your data and weight 0 if $\alpha_i < 0$.
3. **The information ratio** from the Fundamental Law, what it adds to the Sharpe Ratio of the SAA, and how credible it is.

§6 Index model with the SAA as the index

$$R^e_{i,t} = a_i + \beta_i R^e_{S,t} + \varepsilon_{i,t}$$

For each stock, regress its monthly excess returns (last 60 months) on the excess returns of your SAA, the "index" of Round 2. From the regression keep two numbers:

- β_i: how much the stock moves with the SAA, its **systematic risk** relative to your benchmark;
- $\sigma^2(\varepsilon_i)$: the variance of the residuals, annualized ($\times 12$). It is the stock's **idiosyncratic risk**, the part the SAA doesn't explain. It is the risk you take on purpose with the bet, and it enters the Treynor-Black weights in §7.

> **Careful:** a_i is a *historical* average: it is **not** your alpha.

History is used only for risk. Past returns tell you how risky a stock is (β_i, $\sigma^2(\varepsilon_i)$, TE, vol, VaR), because risk is fairly stable over time. They don't tell you its future alpha: that comes from your valuation and catalyst (§4–§5). A discretionary pick can't be backtested.

§7 Treynor-Black

$$w_i \propto \frac{\alpha_i}{\sigma^2(\varepsilon_i)}, \qquad w_i = 0 \ \text{ if } \alpha_i < 0$$

$$\alpha_A = \sum_i w_i \alpha_i, \qquad \beta_A = \sum_i w_i \beta_i, \qquad \sigma^2(\varepsilon_A) = \sum_i w_i^2\, \sigma^2(\varepsilon_i)$$

$$w^0_A = \frac{\alpha_A/\sigma^2(\varepsilon_A)}{E(R^e_S)/\sigma^2_S}, \qquad w^*_A = \frac{w^0_A}{1 + (1-\beta_A)\, w^0_A}, \qquad w^*_S = 1 - w^*_A$$

New portfolio $= (1 - w_A) \cdot \text{SAA} + w_A \cdot A$.

The steps, as in the notebook:

1. Take the **annualized expected alpha** of §5 (valuation $\times$ catalyst), not the historical a_i of §6.
2. Score each stock with $\alpha_i/\sigma^2(\varepsilon_i)$: more alpha per unit of idiosyncratic risk gets more weight. Negative scores are set to 0 (long-only), then the scores are scaled to sum to 1.
3. Compute α_A, β_A, $\sigma^2(\varepsilon_A)$ of the active portfolio and its size w^*_A.

Results of the TA example

$$\alpha_A = 4.35\%, \qquad \beta_A = 0.47, \qquad \sigma(\varepsilon_A) = 21.1\%, \qquad w^0_A = 0.30, \qquad w^*_A = 0.26$$

Only GIS has a positive alpha, so A = 100% GIS, while NKE and HD get 0 (long-only). Treynor-Black wants 26% in GIS and 74% in the SAA. Checkpoint 1 asks for 3–6 stocks, so a group in this situation must replace NKE and HD.

> **Careful:** with $\alpha^{\text{closed}} = 151\%$ for GIS, Treynor-Black gives $w^*_A = 1.59$: 159% in GIS and -59% in the SAA, an absurd position. Huge alphas give huge bets: check the alphas first.

§8 Tracking-error budget

The tracking error is the volatility of the *difference* between your portfolio's return and the SAA's return, annualized:

$$TE = \text{sd}(R_{new} - R_S)\sqrt{12}$$

It measures how far, in a typical year, your result can drift away from the benchmark. TE = 0 means you hold exactly the SAA. TE = 4% means that in a normal year your return can be roughly 4 points above or below the SAA. It is the risk of your *bet*, not the total risk. Round 2 asks for $2\% \leq TE \leq 5\%$: enough active risk to show a view, not so much that the SAA is lost.

The new portfolio is $(1 - w_A) \cdot \text{SAA} + w_A \cdot A$, so the difference from the SAA is

$$R_{new} - R_S = w_A\,(R_A - R_S) \quad \Longrightarrow \quad TE = w_A \cdot \underbrace{\text{sd}(R_A - R_S)\sqrt{12}}_{\text{TE per unit of } A}$$

The TE grows in proportion to w_A: double the bet, double the TE. So the budget turns into a lower and an upper bound on the size:

$$w_A^{lo} = \frac{2\%}{\text{sd}(R_A - R_S)\sqrt{12}}, \qquad w_A^{hi} = \frac{4\%}{\text{sd}(R_A - R_S)\sqrt{12}}$$

We use 4% instead of 5% to stay safely inside the budget: the TE measured during the challenge will not be exactly the ex-ante one.

$$w_A = \min\left(\max(w_A^*,\, w_A^{lo}),\, w_A^{hi}\right)$$

In words: keep the Treynor-Black size if it is inside the budget, otherwise push it to the nearest bound.

The steps, as in the notebook:

1. Monthly return of the active portfolio A with the §7 weights, and its TE per unit, $\text{sd}(R_A - R_S)\sqrt{12}$.
2. The bounds w_A^{lo} (TE = 2%) and w_A^{hi} (TE = 4%), then w_A = the Treynor-Black w_A^* pushed inside them.
3. The new portfolio's returns, its TE, volatility and 1-month 5% VaR next to those of the SAA.

Results of the TA example

$$w_A^* = 26\%, \qquad w_A^{lo} = 9.27\%, \qquad w_A^{hi} = 18.55\% \quad \Longrightarrow \quad w_A = 18.55\%$$

	SAA	New
Ex-ante vol	7.93%	8.15%
1-month 5% VaR	3.77%	3.87%
Tracking error	0	4.00%

Treynor-Black asks for 26%, above the budget, so the TE caps the stocks at 18.5% of the portfolio. Total risk barely moves (vol from 7.93% to 8.15%): the stock bet adds active risk relative to the SAA, not much total risk.

Report the volatility and VaR of the new portfolio **next to those of the SAA**.

§9 Information ratio and the Fundamental Law

IR of each stock and of the active portfolio (Class 3 Part II, eq. 15):

$$IR_i = \frac{\alpha_i}{\sigma(\varepsilon_i)}, \qquad IR_A = \frac{\alpha_A}{\sigma(\varepsilon_A)}, \qquad IR_A^2 = \sum_{i \in A} IR_i^2$$

The sum runs only over the stocks you hold ($\alpha_i > 0$): the negative IR_i of the stocks with weight 0 do not enter.

What it adds to the SAA (eq. 16):

$$SR_P^2 = SR_S^2 + IR_A^2$$

This holds only at the optimal size w_A^*. With the TE budget you hold less ($w_A < w_A^*$), so compute the expected Sharpe at the size you actually hold:

$$E(R^e_{new}) = E(R^e_S) + w_A\left[\alpha_A + (\beta_A - 1)\,E(R^e_S)\right], \qquad SR_{new} = \frac{E(R^e_{new})}{\sigma_{new}}$$

The Fundamental Law (Grinold 1989). To raise IR_A a manager has two levers: **skill** (better forecasts of the alphas) and **breadth** (more bets N). With equal IRs:

$$IR_A = \sqrt{N} \cdot IR_i$$

Example from class: 4 bets with IR 0.25 each give $\sqrt{4} \times 0.25 = 0.50$, and so do 400 bets with IR 0.025. The first is a discretionary fund (Buffett), the second a quant fund (AQR).

Credibility: compare your IR_A with these benchmarks: 0.50 for 4 good bets, 0.64 for Buffett over four decades. An IR_A far above them means your alphas are too optimistic: go back to the valuation inputs and the catalyst.

The steps, as in the notebook:

1. Table of Checkpoint 1: for each stock the alpha, $\sigma^2(\varepsilon_i)$, the weight in A and IR_i.
2. IR_A (eq. 15) and the Sharpe Ratio at w_A^* (eq. 16).
3. The expected Sharpe at the w_A we actually hold (§8).

Results of the TA example

$$IR_A = 0.21, \qquad SR_P = \sqrt{0.26^2 + 0.21^2} = 0.33 \;\; (\text{at } w_A^*), \qquad \text{Sharpe at } w_A = 18.5\% : \; 0.33$$

A realistic gain for one bet with IR 0.21, below the benchmarks (0.50 for 4 bets, Buffett 0.64). Capping the size costs little (0.325 at w_A vs 0.330 at w_A^*). With $\alpha^{\text{closed}} = 151\%$ for GIS we would have $IR_A = 7.12$ and $SR_P = 7.12$: if IR_A is far above 0.5–0.64, the problem is the alphas, not the formula.

Our case, step by step: GIS only, $\alpha = 4.35\% \to w_A^* = 26\% \to$ TE budget allows 9.3%–18.5% $\to w_A = 18.5\%$ $\to$ 81.5% SAA + 18.5% GIS, TE 4.0% $\to$ Sharpe 0.26 $\to$ 0.33.

§10 From weights to Bloomberg

The SAA has J ETFs with weights w_j^{SAA} (including any cash of the SAA), the active portfolio A has N stocks with Treynor-Black weights w_i (summing to 1). Keep $(1 - w_A)$ in the SAA and put w_A in A:

$$w_j^{new} = (1 - w_A)\,w_j^{SAA} \quad (j = 1, \ldots, J), \qquad w_i^{new} = w_A\,w_i \quad (i = 1, \ldots, N)$$

Every ETF is cut by the same factor, so the **SAA mix does not change**, only its size. Example: $w_A = 20\%$ and two stocks with $w_1 = 60\%$, $w_2 = 40\% \to$ each ETF $\times\, 0.8$ (TIP 20% $\to$ 16%), stock 1 gets 12%, stock 2 gets 8%.

$$\text{shares} = \left\lfloor \frac{w^{new} \times \text{NAV}}{P} \right\rfloor$$

Shares are rounded down; what is left stays in cash. In the notebook: each ETF $\times$ 0.815, GIS 18.5% (550,369 shares), cash 1.1%.

> **Careful:** NAV is the value of **your** portfolio at the close of 6 Nov (`PORT`), not the initial 100M. Use the closing prices of Fri 6 Nov. Update the **existing** portfolio with `PRTU` on 7–8 Nov. The report is due **Sun 8 Nov, 23:59**.